Convergence bounds that distinguish Fanions: rank, step budget, curvature, and noise

Research attempt for the Neon repository

21 October 2026

Abstract

Equal exponents in worst-case convergence bounds do not imply equal finite iteration performance. We derive two loss lower bounds for exactly the quadratic class used in the manuscript: one for the rank of the accumulated update, and a stronger one for its accumulated nuclear norm. The latter gives a necessary iteration count proportional to $1/k$ at the same learning rate, independently of momentum. On a fresh 500×500 draw from the specified distribution, Neon requires at least 65,172 iterations at learning rate 0.007 to reach loss 0.001. This is an analytic lower bound, not a training run. We also give convergence upper bounds retaining the singular spectrum, curvature, and oracle error, and quantify the robustness of F-Muon and S-Muon. A saved-log audit supports the Muon–Neon spectral comparison but also falsifies the sufficiency of a Frobenius-only explanation for all methods. These results explain a concrete mechanism behind Neon's poor performance; they do not establish the complete CIFAR/LLM accuracy ordering.

What is established, and what is still missing

Continuation across all ranks.

The companion `fanion_followup.pdf` extends this first attempt with a unified plain/F/S-Fanion convergence bound, tracking for the actual code's Nesterov convention, stronger singular-budget obstructions, solvable rank flows, and a saved quadratic trajectory audit. It also identifies a different learning-rate schedule for F-Fanion-10 in the current benchmark source. The scalar nuclear lower bounds below remain valid; the follow-up retains additional operator and spectral information to strengthen them.

The strongest result for the manuscript.

Theorem 1 applies to its synthetic objective without assuming gradient accuracy, particular momentum coefficients, or an oracle trajectory. It bounds *loss*, and gives a large obstruction for Neon at the learning rates actually used. A separate example gives a tight r/k iteration separation between Fanion-k and Muon even with arbitrary learning rates.

A conditional explanation for deep learning.

The appropriate comparison is the true gradient's projection onto the update, squared, divided by curvature along that update. The singular spectrum already distinguishes Muon from Neon in most saved CIFAR snapshots. Momentum error, directional curvature, and the effect of weight normalization have not been recorded in those snapshots. Their values are needed for a complete explanation of the actual algorithms. In particular, our Frobenius surrogate incorrectly favors F-Neon over Muon; we report this failure rather than interpreting every bound as evidence for the desired ordering.

The experimental premise needs qualification.

Section 5.1 of `jota_submission/main.tex` reports F-Muon and S-Muon reaching the synthetic loss threshold in 910 and 890 iterations, versus 1060 for Muon. Their real-task performance is close to Muon's. Thus the target is Muon being much better than the low-rank Fanions and comparable to its full-rank hybrids, with possible reversals among the latter.

Status of originality.

The ingredients are classical: the descent lemma, best rank approximation, norm inequalities, and gradient-estimator tracking. The Muon specialization of the descent calculation is already present in the literature. The contribution of this attempt is their explicit specialization to Fanions, the finite learning-rate obstruction for the manuscript's quadratic, the hybrid comparison, and the audit. No claim of priority or a complete literature survey is made.

Literature check as of 21 October 2026

The statement that no relevant theory has improved is too strong.

- Kovalev and Borodich, *Non-Euclidean SGD for Structured Optimization*, v2, 22 June 2026 , obtain improved guarantees under structured Hessian and gradient-noise assumptions. Their Assumption 5 puts the relevant bounding operators in a common subspace with an order-preserving projection and algebraic closure properties. Theorems 1–3 cover momentum, extrapolation, and momentum variance reduction. These are conditional structural improvements; they do not assert a universal ordering of Muon, Neon, and all hybrids. This paper is already in the local bibliography.
- Shen et al., *On the Convergence Analysis of Muon*, v3, 28 July 2026 , retain matrix smoothness and Hessian information. Their proofs explicitly contain the Muon decrease $\|G\|_*^2/(2rL)$ and a spectral-smoothness alternative. A spectrum-dependent Muon comparison is therefore not new by itself. Our use of all k, hybrid directions, and a common Frobenius stopping metric makes the comparison more direct for this manuscript.
- Nagashima and Iiduka, v3, 5 March 2026 , claim sharper iteration rates, including schedules with increasing batches. However, Corollary 3.1 defines $D_1 = \Delta/\eta + C_3/(1-\beta)$: its D_1/T term cannot be treated as uniform when η depends on T. The displayed assertion with $\eta = 1/T$ is contradicted by noiseless scalar $f(x) = x^2/2$, $x_0 = 2$, $\beta = 0$: normalized steps of size $1/T$ leave $x_t \geq 1$ through T steps. The script reproduces this check. We do not rely on those claimed faster rates. Increasing batch size also changes oracle cost.
- Ma et al., *Schattor*, June 2026 , study a Schatten family and state dimension-free stationarity guarantees in their abstract. This is further relevant work after the manuscript's main 2025 references; an abstract claim alone is not a proof of the requested optimizer ordering.
- Kovalev's trust-region framework and Gluon remain relevant foundations. General norm guarantees must retain their geometry and use the same accuracy metric before numerical rates are compared.

In particular, improved constants under structure, faster rates under extra regularity, and a change from iteration count to total gradient evaluations are different assertions. None implies that one norm is best on every smooth objective.

Notation and the common comparison metric

Let $f: \mathbb{R}^{m\times n} \to \mathbb{R}$ be differentiable and bounded below by f_*, $r = \min(m,n)$, $G = \nabla f(X)$, and $G = U\mathrm{diag}(\sigma_1, \ldots, \sigma_r)V^\top$, with decreasing nonnegative singular values. Define

$$S_k(G) = \sum_{i=1}^{k} \sigma_i(G), \qquad D_k(G) = \sum_{i=1}^{k} u_i v_i^\top.$$

We take only positive singular directions if fewer than k are nonzero, and set $D_k(0) = 0$. When $\mathrm{rank} G \geq k$,

$$\langle G, D_k(G)\rangle = S_k(G), \quad \|D_k(G)\|_{\mathrm{F}}^2 = k, \quad \|D_k(G)\|_* = k, \quad \|D_k(G)\|_{\mathrm{op}} = 1. \tag{1}$$

Fanion-k uses $X^+ = X - \gamma D_k(H)$, where H may be a stochastic gradient or a momentum matrix. Neon has $k = 1$; Muon has $k = r$.

All stationarity comparisons below use the *same* $\|\nabla f\|_{\mathrm{F}}$, and all loss comparisons use the same $f - f_*$. Comparing an ϵ-spectral norm gradient for Neon with an ϵ-nuclear norm gradient for Muon is not a common stopping criterion.

For the hybrids write $a = \alpha$, $b = 1 - \alpha$ and retain the manuscript's actual direction coefficient η:

$$D_{F,k}(H) = aD_k(H) + b\frac{H}{\|H\|_{\mathrm{F}}}, \qquad D_{S,k}(H) = aD_k(H) + b\eta\,\mathrm{sign}(H). \tag{2}$$

Positive scalar rescaling of an entire direction is absorbed by the learning rate. Rescaling only one summand changes a hybrid's direction.

A loss obstruction for exactly the manuscript's quadratic

Consider, including a possible shift Z,

$$f(X) = \frac{1}{2}\left\|M^{1/2}(X - Z)N^{1/2}\right\|_{\mathrm{F}}^2, \quad M, N \succcurlyeq 0, \qquad L = \|M\|_{\mathrm{op}}\|N\|_{\mathrm{op}}. \tag{3}$$

The manuscript uses $Z = 0$. Define $E_0 = M^{1/2}(X_0 - Z)N^{1/2}$ and $s_j = \sigma_j(E_0)$. No inverse of M or N is required.

> **Theorem 1** (Rank and nuclear-norm obstructions). *Let $X_{t+1} = X_t - \gamma_t A_t$, where $\mathrm{rank} A_t \leq k$ and $\|A_t\|_* \leq k$, with $\gamma_t \geq 0$. Then, for every T,*
>
> $$f(X_T) \geq \tfrac{1}{2}\textstyle\sum_{j>kT} s_j^2,$$
> $$f(X_T) \geq \tfrac{1}{2r}\left[\|E_0\|_* - \sqrt{L}\,k\textstyle\sum_{t<T}\gamma_t\right]_+^2. \tag{4), (5}$$
>
> *The rank bound only needs the rank assumption. The nuclear bound only needs the nuclear-norm assumption.*

Proof. Write $E_T = E_0 - R_T$, where $R_T = \sum_{t<T}\gamma_t M^{1/2} A_t N^{1/2}$. Rank subadditivity gives $\mathrm{rank} R_T \leq kT$. The best approximation of E_0 by a matrix of rank at most kT in Frobenius norm has squared error $\sum_{j>kT} s_j^2$, proving (4). This follows directly by truncating the SVD and its standard optimality characterization.

Also, $\left\|M^{1/2}A_tN^{1/2}\right\|_* \le \sqrt{L}\left\|A_t\right\|_* \le \sqrt{L}k$. The triangle inequality implies $\left\|E_T\right\|_* \ge [\left\|E_0\right\|_* - \sqrt{L}k\sum_{t<T}\gamma_t]_+$. Finally, $\left\|E_T\right\|_* \le \sqrt{r}\left\|E_T\right\|_{\mathrm{F}}$ and $f(X_T) = \left\|E_T\right\|_{\mathrm{F}}^2/2$. □

Why this handles momentum.

The theorem permits any A_t satisfying the two constraints. Hence it covers exact rank-k Fanion updates from any heavy-ball or approximate Nesterov matrix, noisy gradients, adversarial singular-vector choices, and arbitrary learning-rate schedules. Momentum can change the selected singular directions; it does not increase the rank or nuclear norm of a single rank-k partial isometry.

> **Corollary 2** (Necessary iterations at a constant learning rate). *If $f(X_T) \le \epsilon$ and $\gamma_t = \gamma > 0$, then*
>
> $$T \ge \frac{[\left\|E_0\right\|_* - \sqrt{2r\epsilon}]_+}{\sqrt{L}\,k\gamma}, \qquad \sum_{j>kT} s_j^2 \le 2\epsilon. \tag{6}$$
>
> *More generally, the first inequality bounds $\sum_t \gamma_t$.*

A numerical consequence at the manuscript's scale

We generated a fresh draw with $m = n = 500$, independent uniform eigenvalues of M, N in $(0, 1)$, and Gaussian X_0 with standard deviation 0.1, interpreting $\mathcal{N}(0, 0.01)$ as variance 0.01. Working in the eigenbases preserves the Gaussian distribution and these bounds. This is not the original PyTorch seed or a rerun of the published training curves.

The draw gives $f(X_0) = 337.933502$, $\left\|E_0\right\|_* = 456.207241$, and a necessary approximation rank of 484 to reach $\epsilon = 0.001$. At $\gamma = 0.007$, (6) gives the following integer lower bounds:

k	**1**	**2**	**10**	**100**	**500**
Rank obstruction	484	242	49	5	1
Nuclear-norm obstruction	65,172	32,586	6,518	652	131

These are *necessary* times. The bound 131 for Muon is not an assertion that it reaches the threshold at iteration 131. The point is that a very large slow-Neon obstruction already follows from its update geometry. The learning rate for low-rank Fanions in the manuscript was inherited from Muon; this is exactly the setting where this obstruction is relevant.

If Neon's learning rate is multiplied by r, the nuclear-budget obstruction shrinks accordingly, but the rank obstruction remains. The comparison does not show that Neon has the same slowdown after unrestricted independent tuning.

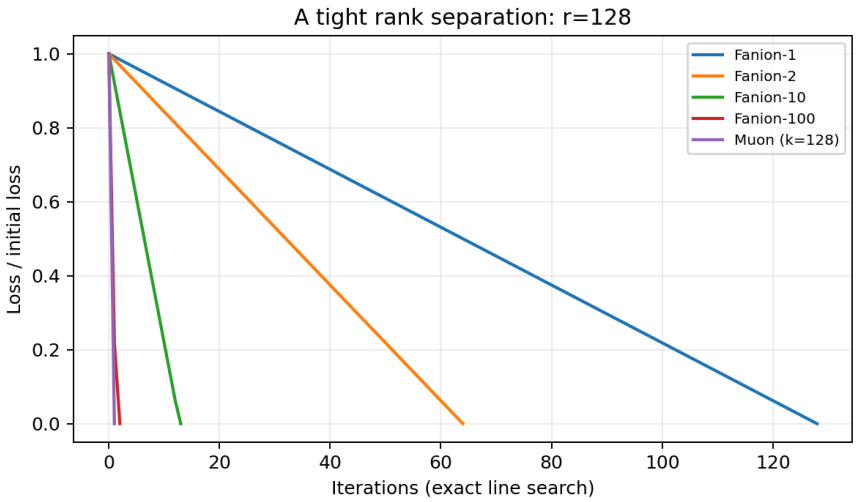
A tight separation after arbitrary step-size tuning

For $f(X) = \left\|X - I_r\right\|_{\mathrm{F}}^2/2$, $X_0 = 0$, every rank-k update method satisfies

$$\frac{f(X_T)}{f(X_0)} \ge \max\{1 - kT/r, 0\}.$$

Fanion-k with exact line search and a diagonal tie choice attains this bound: it fixes k previously uncorrected coordinates in each step (fewer in the last step). Thus it needs exactly $\lceil r/k \rceil$ steps to reach zero, whereas Muon, F-Muon, and S-Muon reach zero in one step because their directions are proportional to I_r on the first iteration.

For arbitrary positive spectra or tie choices this particular attainment claim need not hold; the lower bound still does.



Exact line-search trajectories attaining the rank obstruction. This is an explanatory quadratic, not the manuscript's random instance.

What hybrids change

For $H \neq 0$, the F-Fanion singular vectors are shared by its two summands, so

$$\left\| D_{F,k}(H) \right\|_* = ak + b\frac{\|H\|_*}{\|H\|_{\mathrm{F}}} \leq ak + b\sqrt{r}$$

when rank$H \geq k$. The S-Fanion satisfies

$$\left\| D_{S,k}(H) \right\|_* \leq ak + b\eta\sqrt{r\,mn}\,.$$

Consequently (5) holds for these methods with k in the step budget replaced by these respective upper bounds. Their updates can have full rank, so (4) with the original k is no longer available.

For $r = 500$, $a = b = 1/2$, the F-Neon capacity upper bound is about 11.68, versus 1 for Neon and 500 for Muon. At $\gamma_F = 0.015$, the same new draw requires at least 2,604 F-Neon steps by this bound, versus 117 for F-Muon. The rank floor disappears, but a large nuclear-budget difference remains. The S bound is deliberately conservative because the nuclear norm of a sign matrix depends on its orientation and sign pattern.

Scope with normalization and decay.

The additive-iterate hypothesis holds for the synthetic runs, where `code/L_smooth_tests/run_benchmarks_new.py` sets `norm_weight=False`. If $X_{t+1} = c_t X_t - \gamma_t A_t$, then $X_T = (\prod_t c_t)X_0$ plus a low-rank sum with rescaled coefficients. One must apply the rank comparison to $Z - (\prod_t c_t)X_0$, and account for the products in the nuclear budget. The displayed bounds do not transfer unchanged to CIFAR weight normalization or to arbitrary decay.

Upper bounds retaining the gradient spectrum

Assume Frobenius L-smoothness:

$$f(X+Y) \leq f(X) + \langle G, Y \rangle + \frac{L}{2}\|Y\|_{\mathrm{F}}^2.$$

For a nonzero descent direction D, define

$$q(G,D)=\frac{\langle G,D\rangle^2}{\|G\|_{\mathrm{F}}^2\|D\|_{\mathrm{F}}^2}\in[0,1].$$

(7)

The score is unchanged by positive scalar scaling of D. For an exact Fanion with rank$G\geq k$, it is

$$q_k(G)=\frac{S_k(G)^2}{k\|G\|_{\mathrm{F}}^2}.$$

(8)

For rank-deficient gradients, the same expression remains a conservative quantity; the exact cosine uses $\min(k,\mathrm{rank}G)$ in the denominator.

> **Theorem 3** (Spectrum-dependent descent and PL convergence). *Use the true gradient and set* $\gamma_t=S_k(G_t)/(Lk)$*. Then*
>
> $$f(X_{t+1})\leq f(X_t)-\frac{S_k(G_t)^2}{2Lk}=f(X_t)-\frac{q_k(G_t)}{2L}\|G_t\|_{\mathrm{F}}^2.$$
>
> *(9)*
>
> *In particular,*
>
> $$\sum_{t<T}q_k(G_t)\|G_t\|_{\mathrm{F}}^2\leq 2L(f(X_0)-f_*).$$
>
> *(10)*
>
> *If* $q_k(G_t)\geq \underline{q}_k>0$ *along the trajectory, then* $\min_{t<T}\|G_t\|_{\mathrm{F}}^2\leq 2L(f(X_0)-f_*)/(T\underline{q}_k)$*. If additionally* $\|\nabla f(X)\|_{\mathrm{F}}^2\geq 2\mu(f(X)-f_*)$ *along the trajectory, then*
>
> $$f(X_T)-f_*\leq(1-\mu\underline{q}_k/L)^T(f(X_0)-f_*).$$
>
> *(11)*

Proof. Apply smoothness with $Y=-\gamma_t D_k(G_t)$ and use (1); if the active rank is smaller, $\|D\|_{\mathrm{F}}^2\leq k$ still gives the stated inequality. Substitute the step size, telescope, and then apply the respective lower bounds on q_k and $\|G_t\|_{\mathrm{F}}^2$. □

For a flat nonzero singular spectrum with r modes, $q_k=k/r$. Thus the upper bounds retain a factor r/k in deterministic iterations, and the PL rate is $O((L/\mu)(r/k)\log(\Delta/\epsilon))$ if this flatness condition persists. A more robust sufficient assumption is $\sigma_1/\sigma_r\leq\chi$, which gives $q_k\geq k/(\chi^2 r)$. This is a structural assumption to measure, not a consequence of smoothness.

At a common matrix G, the exact Muon–Neon ratio is

$$\frac{q_r(G)}{q_1(G)}=\frac{1}{r}\left(\frac{\|G\|_*}{\|G\|_{\mathrm{op}}}\right)^2.$$

(12)

The ratio ranges from $1/r$ to r. Muon's Frobenius decrease certificate is larger precisely when $\|G\|_*/\|G\|_{\mathrm{op}}>\sqrt{r}$.

Why a worst-case bound cannot rank them.

In fact,

$$\inf_{G \neq 0} \frac{S_k(G)^2}{k \|G\|_F^2} = \min\{1/k, k/r\}.$$

To see this, express a decreasing nonnegative singular vector as a nonnegative sum of the step vectors $v_j = (1, \ldots, 1, 0, \ldots, 0)$. After imposing $S_k = 1$, it is a convex combination of $v_j/\min(k, j)$. Convexity of the squared Euclidean norm gives its maximum as $\max_j j/\min(k, j)^2 = \max(1, r/k^2)$. Rank-one and flat vectors attain the endpoints. In particular, the worst-case constants for both Neon and Muon are $1/r$. Additional spectrum or curvature information is mathematically necessary.

Upper certificates are not observed speedups.

Equation (12) compares two certified decreases at the *same* point with separately optimized scalar steps. It does not bound the ratio of their complete training times. Different algorithms visit different points; a ratio of upper complexity bounds is not a lower bound on their actual runtime ratio.

F-Muon and S-Muon: explicit robustness bounds

Take $H = G \neq 0$ and $\operatorname{rank} G \geq k$, and put $x = S_k(G)/\|G\|_F$, so that $x^2/k = q_k$.

> **Proposition 4** (Exact F-hybrid formula). *For $a, b \geq 0$, not both zero,*
>
> $$q_{F,k} = \frac{(ax + b)^2}{a^2 k + b^2 + 2abx} \geq q_k. \tag{13}$$

Proof. The numerator follows from $\langle G, D_{F,k} \rangle = aS_k + b\|G\|_F$. The denominator follows from $\langle D_k, G/\|G\|_F \rangle = x$. After multiplying by the positive denominators, the difference from q_k has numerator $(k - x^2)(2abx + b^2) \geq 0$. □

Thus adding a positive normalized *true* gradient can only improve the Frobenius angle. This explains why F-Muon can improve upon Muon on the synthetic task in this particular upper bound. It also proves that this bound alone cannot explain a universal advantage of Muon over NSGD or F-Neon: NSGD has angle one.

> **Proposition 5** (S-hybrid retention). *For $a > 0$, $b \geq 0$, $\eta \geq 0$, and $d_0 = \#\{G_{ij} \neq 0\}$,*
>
> $$q(G, D_{S,k}) \geq \frac{q_k(G)}{(1 + \rho_{S,k})^2}, \qquad \rho_{S,k} = \frac{b\eta\sqrt{d_0}}{a\sqrt{k}}. \tag{14}$$

Proof. $\langle G, D_{S,k} \rangle = aS_k + b\eta\|G\|_1 \geq aS_k$, while $\|D_{S,k}\|_F \leq a\sqrt{k} + b\eta\sqrt{d_0}$. Substitute in (7). No sign assumption on the cross term $\langle D_k, \operatorname{sign}(G) \rangle$ is needed. □

For F-Muon with $a = b = 1/2$, the normalized-gradient component has Frobenius length only $1/\sqrt{r}$ of the Muon component. For full-rank S-Muon on a dense matrix, the corresponding relative length is $\eta\sqrt{\max(m, n)}$. Thus the raw coefficient $\alpha = 1/2$ does not mean equal contribution to the actual update length. For the saved CIFAR shapes and $\eta = 0.003$, (14) preserves between 76.4% and 91.7% of the Muon angle certificate. These are uniform worst-case retentions, not predicted validation accuracies.

More generally, for a PSD curvature majorant $\mathcal{H}$ and $D = aD_0 + bD_1$ with $\langle G, D_1\rangle \geq 0$, the curvature score in the next section retains at least $(1+\rho)^{-2}$ of that of D_0, where $\rho = b\left\|\mathcal{H}^{1/2}D_1\right\|_{\mathrm{F}} / (a\left\|\mathcal{H}^{1/2}D_0\right\|_{\mathrm{F}})$. Here the notation denotes the norm on the vectorized matrix space.

Curvature: the comparison that can account for reversals

Suppose a quadratic upper model is valid along the considered step:

$$f(X - \gamma D) \leq f(X) - \gamma A + \frac{\gamma^2}{2} C(D), \qquad A = \langle G, D\rangle, \quad C(D) > 0. \tag{15}$$

For example, $C(D) = \langle D, \mathcal{H}D\rangle$ if $\nabla^2 f \preccurlyeq \mathcal{H}$ on the segment. The optimized decrease certificate is

$$\frac{1}{2}\mathcal{J}(G, D), \qquad \mathcal{J}(G, D) = \frac{[A]_+^2}{C(D)}, \quad \gamma = \frac{[A]_+}{C(D)}. \tag{16}$$

If the model is only valid up to a maximal step, truncate γ and use (15) directly. For a quadratic, the model is exact at every step, so this is the *actual* optimal one-step decrease.

For (3),

$$C(D) = \langle D, MDN\rangle = \left\|M^{1/2}DN^{1/2}\right\|_{\mathrm{F}}^2. \tag{17}$$

Writing u_i, v_i for the singular vectors of G, the Fanion curvature is

$$C(D_k) = \sum_{i,j \leq k} (u_i^\top M u_j)(v_i^\top N v_j).$$

The cross terms matter. Replacing this by Lk may obscure the correct ordering. In particular, at a common point Muon's curvature score beats Neon's exactly when

$$\left(\frac{\|G\|_*}{\|G\|_{\mathrm{op}}}\right)^2 > \frac{C(D_r)}{C(D_1)}. \tag{18}$$

This is a directly testable condition. Under the Frobenius majorant it reduces to the threshold in (12).

An explicit structured quadratic with the desired ordering

Take $m = n = r, N = I, M = \Lambda = \mathrm{diag}(\lambda_i) \succ 0$, and $X_0 = I$. The optimum of $f(X) = \langle X, \Lambda X\rangle/2$ is zero, and $G_0 = \Lambda$. Muon's direction is I, so step one solves the problem. Neon with exact line search only removes the diagonal entry with the largest remaining λ_i, and requires exactly r steps for zero loss. Fanion-k takes $\lceil r/k \rceil$. S-Muon is proportional to I on this iterate, and likewise solves the problem in one step.

For F-Muon, choose step $1/a$ rather than even optimizing it. Then

$$\frac{f(X_1)}{f(X_0)} \leq \left(\frac{b}{a}\right)^2 \frac{\sum_i \lambda_i^3}{(\sum_i \lambda_i^2)(\sum_i \lambda_i)} \leq \left(\frac{b}{a}\right)^2 \frac{\kappa}{r}, \quad \kappa = \frac{\lambda_{\max}}{\lambda_{\min}}.$$

Hence, for bounded κ, F-Muon is close to the one-step optimum as r increases. This is a member of the manuscript's quadratic class with a structured initialization, not a claim about all Gaussian initializations.

As a numerical illustration, for $r = 128$ and log-spaced $\lambda_i \in [0.001, 1]$, exact line search gives:

Direction	Frobenius angle q	Remaining fraction of loss
Neon	0.1031	0.947011
F-Neon	0.6605	0.614410
NSGD	1.0000	0.249105
Muon	0.2868	0
F-Muon	0.3531	0.006614
S-Muon	0.2868	0

The curvature-aware calculation recovers the ordering even though the Frobenius angles favor NSGD and F-Neon. This illustrates what information is missing from a norm-only explanation.

A counterexample to a universal Muon advantage

For $f(X) = \|X\|_{\mathrm{F}}^2/2$ at $X = \mathrm{diag}(1, \delta, \ldots, \delta)$ with $\delta > 0$, both directions have full-rank gradient inputs, and exact line search gives

$$\frac{\text{Muon one-step decrease}}{\text{Neon one-step decrease}} = \frac{(1 + (r-1)\delta)^2}{r}.$$

At $r = 64$, $\delta = 0.001$, Neon has 56.64 times the one-step decrease. Consequently smoothness, strong convexity, and exact gradients alone do not imply the desired local ordering. A theorem imposing it in all cases would be false.

Stochastic gradients, momentum, and inexact LMOs

The next result keeps the estimator error visible instead of assuming orthogonalization of a stochastic matrix is unbiased.

> **Theorem 6** (Oracle-input error). *Let $H = G + E$, and let $\tilde{D}$ satisfy*
>
> $$\left\|\tilde{D}\right\|_{\mathrm{op}} \le 1, \quad \left\|\tilde{D}\right\|_{*} \le k, \qquad \delta = S_k(H) - \langle H, \tilde{D}\rangle \ge 0.$$
>
> *Then*
>
> $$\langle G, \tilde{D}\rangle \ge S_k(G) - 2S_k(E) - \delta, \qquad \left\|\tilde{D}\right\|_{\mathrm{F}}^2 \le k. \tag{19}$$
>
> *Thus an L-smooth objective obeys*
>
> $$f(X - \gamma\tilde{D}) \le f(X) - \gamma S_k(G) + 2\gamma S_k(E) + \gamma\delta + L\gamma^2 k/2. \tag{20}$$

Proof. The support function of $\{D: \|D\|_{\mathrm{op}} \le 1, \|D\|_{*} \le k\}$ is S_k. Thus $\langle G, \tilde{D}\rangle \ge S_k(H) - \delta - S_k(E)$. Use the triangle inequality $S_k(H) \ge S_k(G) - S_k(E)$. Also $\left\|\tilde{D}\right\|_{\mathrm{F}}^2 \le \left\|\tilde{D}\right\|_{\mathrm{op}}\left\|\tilde{D}\right\|_{*} \le k$. Apply smoothness for the last

assertion. □

For an arbitrary momentum matrix, take H to be any positive rescaling of the *actual* oracle input; $E = H - G$ then contains both noise and tracking bias. For an approximate Newton–Schulz update outside the feasible set, the feasibility and $\left\|\tilde{D}\right\|_{\mathrm{F}}^2 \le k$ claims cannot be assumed: use the actual direction and its norm in (15), or control its violation separately.

An explicit sample bound with minibatches

Use the normalized Frobenius step $X_{t+1} = X_t - hD_k(H_t)/\sqrt{k}$. Suppose $q_k(G_t) \ge \underline{q} > 0$ almost surely and

$$\mathbb{E}S_k(H_t - G_t)/\sqrt{k} \le \nu_k/\sqrt{B},$$

with exact LMOs. Writing $\Delta = f(X_0) - f_*$, telescoping (20) yields

$$\frac{1}{T}\sum_{t<T}\mathbb{E}\left\|G_t\right\|_{\mathrm{F}} \le \frac{1}{\sqrt{\underline{q}}}\left(\frac{\Delta}{hT} + \frac{Lh}{2} + \frac{2\nu_k}{\sqrt{B}}\right). \tag{21}$$

For $h = \sqrt{2\Delta/(LT)}$, it suffices to choose

$$T \ge \frac{8L\Delta}{\underline{q}\epsilon^2}, \qquad B \ge \frac{16\nu_k^2}{\underline{q}\epsilon^2}$$

to make the left side at most ϵ. Rounding to positive integers gives total gradient evaluations of order

$$\left(1 + \frac{L\Delta}{\underline{q}\epsilon^2}\right)\left(1 + \frac{\nu_k^2}{\underline{q}\epsilon^2}\right).$$

The exponent four in the noise-dominated term remains, but its coefficient retains $1/\underline{q}^2$ and the projected noise ν_k^2. Under a common Frobenius variance bound $\mathbb{E}\left\|H_t - G_t\right\|_{\mathrm{F}}^2 \le \sigma^2/B$, one may conservatively take $\nu_k = \sigma$. The flat-spectrum condition then gives rank factors r/k and $(r/k)^2$ in these sufficient iteration and sample bounds. Those are comparisons of certificates under the stated trajectory conditions, not proved time ratios between arbitrary stochastic training runs.

A fixed-batch momentum bound with the same spectral factor

For clarity, this calculation is for EMA heavy ball, not the implemented Nesterov variant. Let $m_{-1} = 0$, $\alpha = 1 - \beta \in (0, 1]$, $m_t = \beta m_{t-1} + \alpha g_t$, and $X_{t+1} = X_t - hD_k(m_t)/\sqrt{k}$. Assume conditionally unbiased $g_t = G_t + \xi_t$ with conditional Frobenius variance at most σ^2, and $q_k(G_t) \ge \underline{q}$. Then

$$\frac{1}{T}\sum_{t<T}\mathbb{E}\left\|G_t\right\|_{\mathrm{F}} \le \frac{1}{\sqrt{\underline{q}}}\left(\frac{\Delta}{hT} + \frac{Lh}{2} + \frac{2\beta\left\|G_0\right\|_{\mathrm{F}}}{\alpha T} + \frac{2\beta Lh}{\alpha} + 2\sigma\sqrt{\frac{\alpha}{2-\alpha}}\right). \tag{22}$$

To verify this, unroll

$$m_t - G_t = -\beta^{t+1}G_0 + \sum_{s=1}^{t}\beta^{t-s+1}(G_{s-1} - G_s) + \alpha\sum_{s=0}^{t}\beta^{t-s}\xi_s.$$

The drift norm is at most $\beta Lh/\alpha$ because each iterate moves by at most h. Martingale orthogonality bounds the last term's expected norm by $\sigma\sqrt{\alpha/(2-\alpha)}$. Average the initial term, apply Theorem 6, and telescope.

Taking α of order $\min(1, \underline{q}\epsilon^2/\sigma^2)$ and h of order $\epsilon\sqrt{\underline{q}}\alpha/L$, with sufficiently small constants, gives the sufficient complexity

$$T = O\left(1 + \frac{L\Delta}{\underline{q}\epsilon^2} + \frac{L\Delta\sigma^2}{\underline{q}^2\epsilon^4} + \frac{\|G_0\|_{\mathrm{F}}}{\sqrt{\underline{q}}\epsilon} + \frac{\|G_0\|_{\mathrm{F}}\sigma^2}{\underline{q}^{3/2}\epsilon^3}\right). \tag{23}$$

This displays the finite-precision terms as well as the fourth-power asymptotic term. It is a straightforward specialization of standard tracking arguments, not a claim to improve the universal stochastic first-order exponent.

Audit of the repository's CIFAR gradient logs

The script analyzes the predeclared cohort `code/cifar_norms/grad_norms_*_run0_20251208_*.csv`: nine trajectories, all 250 logged steps in each, all six trained convolution matrices, totaling 13,500 layer snapshots. Shapes are inferred from `CifarNet` in `code/cifar_bench/norm_logger_airbench.py`: 64×216, 64×576, 256×576, and three 256×2304 matrices. The CSVs themselves do not store shape metadata.

The following table uses the Muon trajectory as a common collection of matrices, not a comparison of different stopping norms. The same diagnostic is also output for all other trajectories.

Layer	r	**median** q_r/q_1	$\Pr(q_r > q_1)$	**median** $q_{F,r}/q_r$	**median** $q_{F,1}/q_r$
1.conv1	64	1.67	88.0%	1.17	1.62
1.conv2	64	2.84	98.4%	1.10	1.13
2.conv1	256	2.81	97.2%	1.10	1.55
2.conv2	256	3.81	97.2%	1.07	1.25
3.conv1	256	1.94	89.6%	1.10	1.63
3.conv2	256	2.10	96.0%	1.14	2.08

Here Pr denotes the fraction of logged snapshots, not a statistical population probability or a confidence interval. $a = b = 1/2$ for both F columns. The q values are computed from the three saved norms using (8) and (13).

Positive finding.

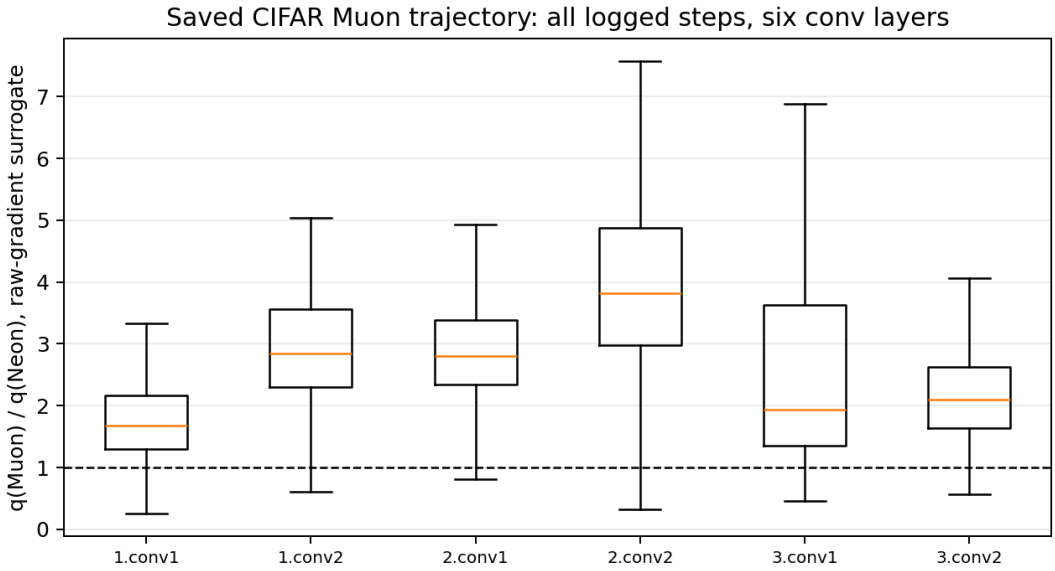
The median Muon–Neon ratio is between 1.67 and 3.81 across the six layers, and favors Muon in 88.0–98.4% of snapshots. Hence the observed spectra are broadly compatible with a better Muon descent certificate.

Negative finding that limits the conclusion.

The F-Neon–Muon surrogate ratio has median 1.13–2.08, favoring F-Neon. Yet F-Neon is empirically weaker. Therefore these spectra and a common Frobenius smoothness constant do *not* suffice to explain the complete ranking. Curvature, actual momentum alignment, and finite learning-rate budgets must be retained. F-Neon also has a much smaller update length at the manuscript's inherited learning rate than an independently rescaled angle comparison permits.

Data limitations.

The CSVs store raw minibatch gradient norms, not the full population gradient, the momentum input, exact singular vectors, Hessian products, or update matrices. A nonlinear map of a stochastic gradient is not generally an unbiased map of the true gradient. Moreover, the reference Muon file has only 8 finite loss entries out of 250, although all the norm entries used here are finite. We do not fit loss rates to that column, infer the cause of its nonfinite entries, or use this cohort's final accuracy as the manuscript's eight-epoch result. Source filenames, hashes, counts, and finite-loss counts are saved in `results/summary.json`.



Muon–Neon Frobenius certificate ratios from the same saved raw gradient matrices. Boxes summarize the reference trajectory; they are not measured speedups or convergence rates.

How to turn the conditional part into an experimental explanation

The missing measurements are specific and considerably smaller than a full Hessian. On common saved checkpoints and fixed evaluation batches:

1. Save the actual oracle input H_t and the actual update matrix for each method, including Newton–Schulz approximation and scaling. Use a shared large-batch evaluation gradient G_t to measure $\langle G_t, D_t \rangle$ and the error $H_t - G_t$ after an appropriate positive rescaling of H_t.
2. Measure $\langle D_t, \nabla^2 f(X_t) D_t \rangle$ with one Hessian–vector product per candidate direction, or use central loss differences at several small steps. On a nonconvex neural network a pointwise Hessian value is only a local estimate; verify the quadratic upper model at the actual step, rather than calling that value a global smoothness constant.
3. Compare $\gamma_t \langle G_t, D_t \rangle - \gamma_t^2 C_t(D_t)/2$ at the actual learning rate and $\mathcal{I}(G_t, D_t)$ at a separately optimized step. This separates a finite step-budget limitation from a directional curvature limitation. With multiple layers, include cross-layer curvature or justify a block-diagonal upper model before adding per-layer certificates.
4. For the quadratic, compute E_0 from the original saved M, N, X_0 and substitute its singular values into Theorem 1. This yields a bound on the original run without a new optimizer sweep.

`analyze.py --snapshot example.npz` already computes the exact projections and, when the NPZ contains `left` and `right`, the exact quadratic curvature scores. It accepts `oracle_input` separately from `gradient`; therefore it does not silently equate the raw gradient with a Nesterov direction.

Implementation and manuscript details discovered during the check

These observations are recorded here; no original source was edited.

1. Equation `eq:ckfk_norm` in the manuscript puts $(1-\alpha)/\eta$ before the entrywise ℓ_1 norm, while its update uses $(1-\alpha)\eta\mathrm{sign}(H)$. The support function of the actual Minkowski-sum ball is $\alpha S_k(H) + (1-\alpha)\eta\|H\|_1$. Thus that equation's coefficient should be $(1-\alpha)\eta$ to match the displayed update.
2. The manuscript defines a normalized EMA buffer $B_t = \beta B_{t-1} + (1-\beta)g_t$ and approximate Nesterov input $g_t + \beta B_t$. `code/optimizers.py` accumulates $b_t = \beta b_{t-1} + g_t$ and uses $g_t + \beta b_t$. With $B_t = (1-\beta)b_t$, the code input is $g_t + \beta B_t/(1-\beta)$. The two mixtures are generally not positive scalar multiples, unlike the heavy-ball case. Theory should specify which one is being analyzed.
3. The Frobenius smoothness constant of the quadratic is $L = \|M\|_{\mathrm{op}}\|N\|_{\mathrm{op}}$. Some helper expressions in `code/L_smooth_tests/problems.py` use other, looser quantities. The obstruction proved here uses the exact Frobenius constant and does not rely on those helper expressions.

Reproduction and conclusion

From the repository root, run

```
python theory_attempts/analyze.py
python theory_attempts/build_report.py
```

The supplied PDF and standalone HTML use Pandoc's MathML export and a headless browser, keeping browser profiles in this folder. This avoids the current sandbox's MiKTeX initialization failure. For a normal LaTeX build instead, use:

```
cd theory_attempts
pdflatex -interaction=nonstopmode -halt-on-error bounds.tex
pdflatex -interaction=nonstopmode -halt-on-error bounds.tex
```

The Python script uses NumPy and Matplotlib on CPU and writes only to `theory_attempts/results`. The run checked 160 random rectangular cases for each angle/oracle assertion, 800 exact quadratic decrease identities, and 640 trajectory points for each quadratic lower bound. The largest relative error in the quadratic identities was below $1.5 \cdot 10^{-15}$. These numerical checks complement the proofs; they do not prove theorems or validate assumptions on neural-network trajectories.

Answer to the research question.

Yes, there are substantially more discriminating bounds than a common $O(\epsilon^{-4})$ statement: a rigorous finite-step explanation of Neon's slow progress on the synthetic class, a tight r/k separation example, and spectrum/curvature/noise-dependent upper bounds with explicit hybrid robustness. The quadratic obstruction is ready to use after substituting the original instance's spectrum. A complete quantitative explanation of the CIFAR/LLM ordering remains open in this attempt because the required curvature and momentum measurements are absent. The existing log audit shows exactly where a simpler explanation succeeds and where it fails.

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